Nuclear Instruments and Methods in Physics Research A 607 (2009) 394–397

Contents lists available at ScienceDirect

Nuclear Instruments and Methods in Physics Research A

journal homepage: www.elsevier.com/locate/nima

Micro-Drift Chamber as a precise vertex detector for the DIRAC experiment

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ARTICLE INFO

Article history:
Received 19 December 2008
Received in revised form
21 May 2009
Accepted 3 June 2009
Available online 11 June 2009

Keywords:
Vertex detector
Drift chamber
Multiwire chamber

ABSTRACT

A possible use of the Micro-Drift Chamber (MDC) as a high-resolution vertex detector in the upstream part of the DIRAC spectrometer was investigated. Simulations of different MDC layouts were performed with the help of the GARFIELD package. On the basis of the simulation results the optimal chamber geometry and gas mixture were selected.

A one-cell prototype was produced and its characteristics were measured at different particle fluxes, various gas pressures and electronics thresholds.

The observed data together with the expected MDC coordinate resolution allow this detector to be used in various fields, especially in the DIRAC experiment.

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1. Introduction

The DIRAC experiment [1] aims to measure with high precision the lifetime of $\pi^+\pi^-$ atoms as well as atoms consisting of charged π and K-mesons [2]. The experiment is under way at the CERN Proton Synchrotron on the extracted 24 GeV/c proton beam. The DIRAC setup (Fig. 1) [3] is a magnetic double-arm spectrometer. It consists of a proton beam line, target station, secondary particle channel, spectrometer magnet, upstream and downstream detectors including 28 planes of drift chambers as the main tracking system.

Events investigated in the experiment are characterized by pairs of particles with small relative momentum ($Q<3$ MeV/c). In the upstream part of the spectrometer the distance between such particles is ~200 μm. The MSGC, SFD and IH detectors are used for their detection. But the absence of a rigorous tracking system, which is able to register reliably close tracks, affects the experimental efficiency. Therefore, it seems quite reasonable to supplement the experimental setup with such a system.

Traditional vertex coordinate detectors used for registration of the events with close tracks contain a large number of channels. Therefore, they are very delicate and rather expensive. The amount of information from these detectors is also extremely large and complicates data acquisition and analysis. Widely used solid-state and gaseous multi-channel vertex detectors are universal instruments appropriate for experiments with high particle multiplicity. However, there are many experiments where multiplicity is not so high, but it is necessary to detect pairs of close particles and to measure their coordinates with a high accuracy. DIRAC belongs to this kind of experiment.

The purpose of the present work is to investigate the possibility of developing a vertex detector based on the conventional technique of the drift chamber with a small drift distance—a Micro-Drift Chamber (MDC).

Below we describe the operation of this detector, which thoroughly meets the requirements of DIRAC and similar experiments. Results of the detector property simulation using the GARFIELD package are presented. Some of the data obtained in the experimental investigation of the one-cell MDC prototype are shown.

2. Registration features for two close tracks in a drift chamber

Let us consider how particles are detected in a drift chamber cell. In the case of one particle, primary ionization electrons drift towards the anode wire and initiate the avalanche process. This avalanche occupies part of the anode wire and inhibits for some time the detection of another particle close in space. Therefore, if two particles cross the cell simultaneously, one of them could not be detected either because it does not produce an avalanche or due to electronics dead time.

These restrictions are important in the case of using one plane of the drift chamber. A complementary plane shifted by half a cell width, which is usually employed to eliminate the left–right ambiguity, allows to resolve the problem of detecting two close tracks (Fig. 2). As shown in the figure, if two tracks cross one cell, the signal in each plane will be produced by that particle which is closer to the wire. Therefore, both particles will be registered.

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[1] Supported by INTAS YSF Ref. no. 05-109-5080.

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doi:10.1016/j.nima.2009.06.005

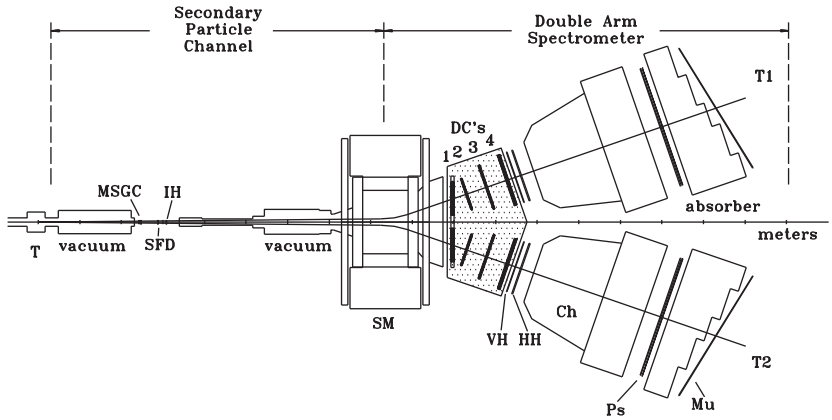

Fig. 1. Experimental setup of the DIRAC experiment: T—target station, MSGC—micro-strip gas chambers, SFD—scintillating fiber detector, IH—ionization hodoscopes, SM—spectrometer magnet, DC—drift chamber system, VH—vertical hodoscopes, HH—horizontal hodoscopes, Ch—Cherenkov counters, Ps—preshower detectors, Mu—muon scintillation counters, T1 and T2—spectrometer arms.

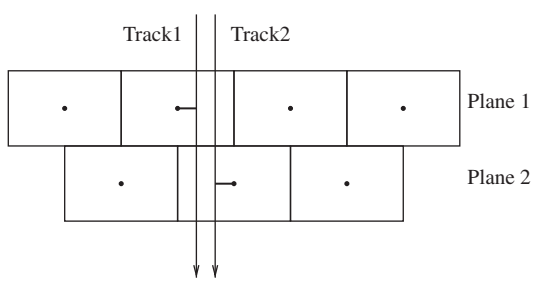

Fig. 2. Scheme of close tracks registration by two shifted drift chamber planes.

In addition to the possibility of detecting two close tracks, the double-plane chamber also provides an interesting opportunity to organize a fast and effective trigger for selection of such events. Crossing two planes, a particle hits two overlapping cells (one in each plane), and the sum of the drift times for these cells is equal to the maximum drift time. Here we make two assumptions which have an irrelevant influence on the trigger efficiency: the track is perpendicular to the chamber, and the drift time is proportional to the drift distance.

In the case of two particles this sum decreases with increasing distance between these tracks. If we feed the signals from these cells into the unit which operates as a mean-timer, we get a time spectrum with the peak corresponding to the case of one particle and the rest of the spectrum corresponding to more than one particle. Then, if we pass the signals from the mean-timers through the time window, we can select the required events in flight. The efficiency of this trigger and its possible realization in the DIRAC experiment were reported in Ref. [4].

3. Choice of MDC geometric parameters and operation mode

The lateral dimension of the secondary particle beam in the supposed MDC region in the DIRAC setup is $\sim 80 \times 80\,\mathrm{mm}^2$ and the particle flux is $\sim 10^6\,\mathrm{s}^{-1}\,\mathrm{cm}^{-2}$. It is necessary to minimize the probability of overlaying tracks from successive interactions of the beam particles with the target. Considering that the number of channels in one MDC plane should be a multiple of 16 and the size of one cell should be small enough to distinguish efficiently two close tracks, the distance between the signal wires was chosen to be $s = (1/10)' = 2.54\,\mathrm{mm}$.

In a number of articles [6–9] the fact that a drift chamber with a small drift distance ($\sim 1-3\,\mathrm{mm}$) could operate in a high gas gain mode ($\sim 10^7$) and at a high counting rate ($\sim 10^7\,\mathrm{s}^{-1}\,\mathrm{cm}^{-2}$) is shown. Operation of drift chambers in the high gas gain mode has a number of advantages. A large signal amplitude makes possible the use of high-threshold readout electronics, and short pulses with a fast leading edge improve the time properties of the chamber.

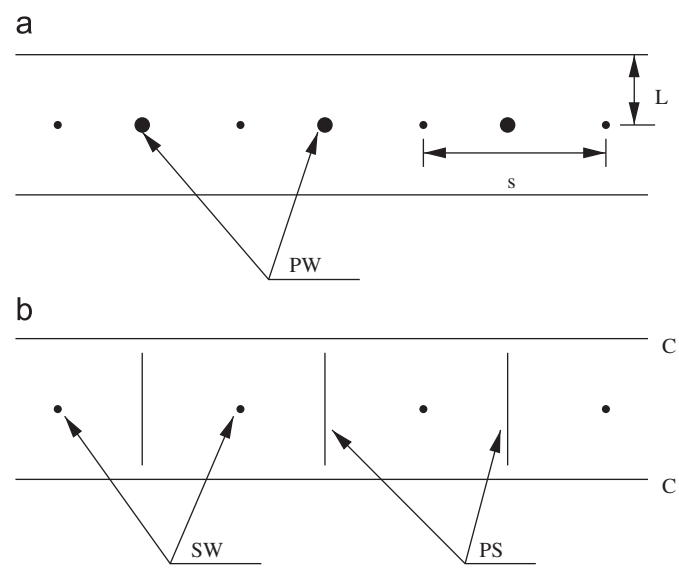

Fig. 3. Schematic layouts of the Micro-Drift Chamber electrodes: with potential wires (a), with potential strips (b). PW—potential wires, $d_{pw} = 100\,\mu\mathrm{m}$ in diameter; SW—anode wires, $d_{sw} = 50\,\mu\mathrm{m}$ in diameter; PS—potential strips, $d_{ps} = 1.6\,\mathrm{mm}$ wide; C—cathodes, Mylar foil, $20\,\mu\mathrm{m}$ thick; $s = 2.54\,\mathrm{mm}$ is the wire spacing; $L = 1\,\mathrm{mm}$ is the gap between the sensitive and cathode planes.

Now we give the analysis of the cell geometry and gas mixture performed with the GARFIELD package [5].

3.1. Drift cell geometry

We consider two possible MDC cell layouts (Fig. 3). It is supposed that anode wires are under zero potential while cathode planes and field-forming electrodes are under the common negative potential.

Fig. 4 shows the electric field map for different drift cell layouts. The cell with potential wires has rather large regions with a low electric field, which deteriorates time properties of the chamber. Another disadvantage of this layout is that the adjacent anode wire may also be hit if the particle crosses the sensitive

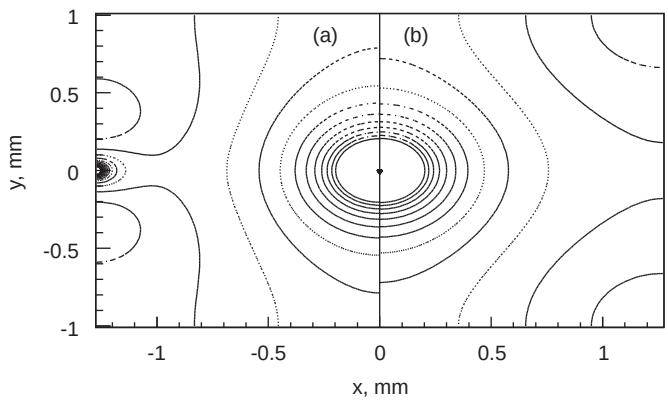

Fig. 4. Electric field map inside different drift cell layouts: (a) cell with potential wires and (b) cell with potential strips, calculated by the GARFIELD package. The chamber voltage is $U = 2.5$ kV. Contours correspond to the field intensities of $2.5, 5.0, \ldots, 30.0$ kV/cm.

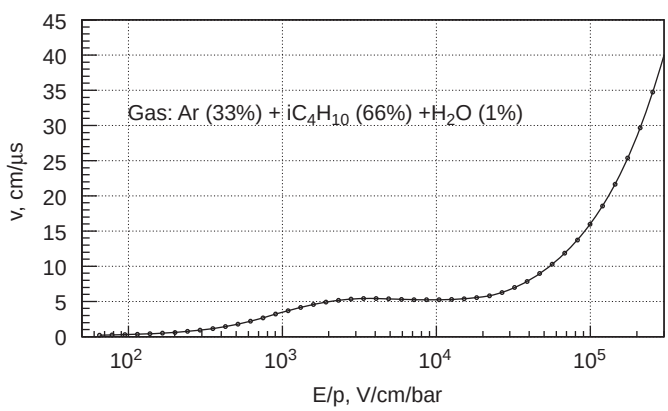

Fig. 5. Dependence of the drift velocity on the electric field, calculated by the GARFIELD package.

plane near the potential wire. This is undesirable in the high counting rate condition.

3.2. Gas mixture

We chose the mixture Ar(33%) + iC_4H_{10}(66%) + H_2O(1%) for the Micro-Drift Chambers. This gas mixture is used in the DIRAC Drift Chamber system. It has the following advantages:

- stable operation in the high gas gain mode (threshold of electronics $I_{th} = 50\,\mu$A);
- wide efficiency plateau (>1 kV);
- fast leading edge time of pulses ($\sim$5 ns);
- weak dependence of the drift velocity v on the electric field E/p practically over the entire field inside the drift cell (Fig. 5) and
- operation at a rather high counting rate.

For this gas mixture at normal pressure, the mean number of primary ionization clusters generated by a charged pion with momentum 2 GeV/c inside the MDC cell is $\sim$17. If particles cross the cell near the anode wire, fluctuations of primary ionization cause a significant drift time spread. This spread could be reduced by increasing the pressure inside the chamber. As will be shown below, an increase in pressure leads to a more stable chamber operation.

Fig. 6 shows the simulation results of the drift function for the selected MDC layout and the gas mixture at the absolute pressure

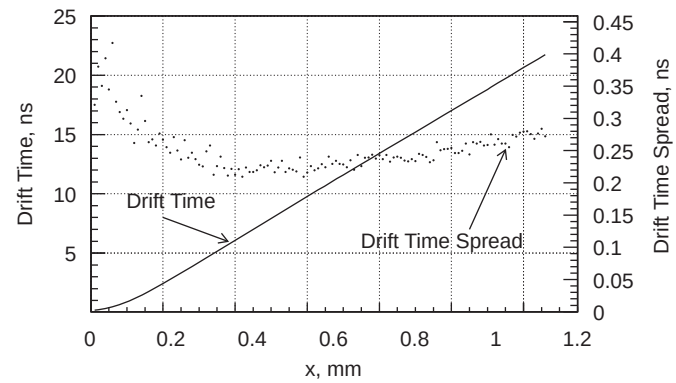

Fig. 6. Drift time and its spread calculated by the GARFIELD package for the MDC at $U = 3.5$ kV.

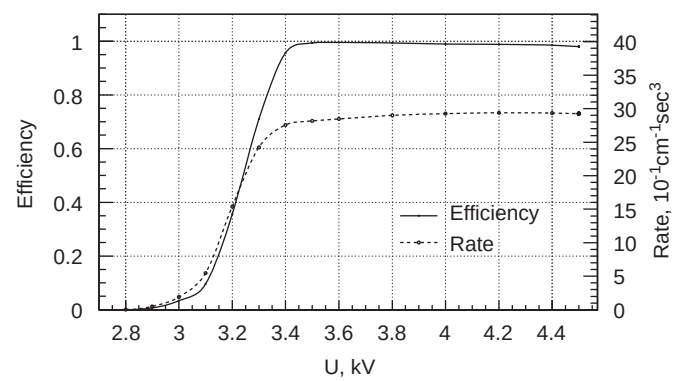

Fig. 7. Efficiency and single counting rate of one MDC cell at the electronics threshold $I_{th} = 10\,\mu$A.

$p = 2$ bar(a). The drift function is almost linear, the maximum drift time is $t_{max} = 22$ ns, and the drift time spread due to primary ionization fluctuations, diffusion and other processes in the gas does not exceed 0.5 ns.

4. One-cell prototype tests of the Micro-Drift Chamber

To study properties of the Micro-Drift Chamber the one-cell prototype with potential strips was made (Fig. 3b). It consists of an anode (signal) beryllium bronze wire of diameter $d_{sw} = 40\,\mu$m, potential strips and cathode planes made of 25 μm thick Mylar foil coated with carbon of $400\,\Omega/\square$ (ohms per square) sheet resistance. The drift cell dimensions are $2.5 \times 2 \times 80\,\text{mm}^3$. The prototype was placed into a pressurized box to vary the pressure inside the cell.

We used a double-counter scintillation telescope and a collimated radioactive ^{90}Sr source to measure chamber characteristics.

The efficiency and the single counting rate curves of the MDC cell were measured at different values of the particle flux, gas pressure and electronics threshold. Time spectra were also measured.

The particle flux was defined by the distance between the radioactive source and the collimator and varied within $(1-50) \times 10^3\,\text{s}^{-1}\,\text{cm}^{-1}$.

The hit wire signals went via the preamplifier to the discriminator, then to the coincidence circuit, where the scintillation counter signals also arrived, and to the counter. The discriminator thresholds for the MDC signal were $I_{th} = 5, 10, 20$ and 50 μA.

The chamber was filled with the $Ar(33\%) + iC_4H_{10}(66\%) + H_2O(1\%)$ gas mixture. The measurements were performed at three values of the absolute gas pressure inside the chamber $p = 1.0, 1.5, 2.0$ bar. We found that the increasing gas pressure led to more stable operation of the detector. The efficiency was extended, the pulse shape became better and the drift time spread was reduced.

Fig. 7 shows the typical efficiency and the single counting rate curves measured at the pressure $p = 2.0$ bar(a) and the discriminator threshold $I_{th} = 10\,\mu A$.

The MDC time spectra were also obtained at different values of the gas pressure, electronics threshold and voltage corresponding to the efficiency plateau for the given threshold. Due to low energy of ^{90}Sr electrons it was impossible to ensure even irradiation of the drift cell. Therefore, the time spectra have a rather complicated shape and are not shown here because their shape does not provide information about the drift function form. The width of the spectra is around 20 ns, which corresponds to the maximum drift time calculated by the GARFIELD package.

5. Conclusion

In this work we have studied the properties of the prototype drift chamber with a small drift distance operating in the high gas gain mode with the view of using such chambers as coordinate detectors in the upstream part of the DIRAC setup.

The optimal chamber layout and gas mixture were chosen on the basis of the experimental requirements and the GARFIELD simulation results. The one-cell drift chamber prototype was constructed, and its characteristics were measured. These measurements were performed using the radioactive ^{90}Sr source at various gas pressures inside the drift cell, discriminator thresholds and particle fluxes through the drift cell.

We found that the chamber operates more stably at increased pressure. At $p = 2$ bar(a), the counting rate per wire unit of length up to $5 \times 10^4\,s^{-1}\,cm^{-1}$ and $50\,\mu A$ threshold, the efficiency plateau width is a few hundred volts.

The obtained data together with the expected coordinate resolution make the Micro-Drift Chamber suitable for a number of applications including the DIRAC experiment.

The next step for the development of a precise vertex detector for the DIRAC experiment is the production of the double-plane prototype, its beam test to check the chamber stability in hard radiation conditions and estimation of the main characteristics (efficiency, coordinate resolution and double-track resolution).

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